

Arvind Ramaswami

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Education

Purdue University

2023–Present

Ph.D. in Computer Science

Advisor: Steve Hanneke

Research areas: Machine learning theory; oracle-efficient online learning

Georgia Institute of Technology

2017–2022

B.S. and M.S. in Computer Science

Advisors: Richard Peng, Sebastian Pokutta, and Jacob Abernethy

B.S. threads: Theory and Intelligence M.S. specialization: Machine Learning

Publications

- [1] I. Attias, S. Hanneke, and **A. Ramaswami**. “Tradeoffs between Mistakes and ERM Oracle Calls in Online and Transductive Online Learning.” *Advances in Neural Information Processing Systems* (NeurIPS), vol. 38, pp. 75639–75680, 2026. **Spotlight**.
- [2] I. Attias, S. Hanneke, and **A. Ramaswami**. “Sample Compression Scheme Reductions.” *Algorithmic Learning Theory* (ALT), PMLR, pp. 134–162, 2025.
- [3] J. Chen, L. Chen, Y. P. Liu, R. Peng, and **A. Ramaswami**. “Exponential Convergence of Sinkhorn under Regularization Scheduling.” *SIAM Conference on Applied and Computational Discrete Algorithms* (ACDA), pp. 180–188, 2023.

Preprints and Submissions

I. Attias, S. Hanneke, and **A. Ramaswami**. “Regret-Oracle Complexity Tradeoffs in Agnostic Online Learning.”
arXiv:2605.07155, 2026.

Work Experience

Mercor

Oct 2024–Jul 2025

ML Intelligence Engineer (Contract)

San Francisco, CA

- Created datasets for OpenAI’s PaperBench.

Genesis Therapeutics

Feb 2023–Aug 2023

ML Research Intern

Burlingame, CA

- Built pose-ranking benchmarks to evaluate ligand pose scoring methods at a machine learning startup focused on drug discovery.
- Developed machine learning-based scoring methods for ligand poses.

PerfectRec

Jun 2022–Aug 2022

ML Engineer Intern

Seattle, WA

- Developed the startup’s first machine learning-powered recommendation system using AWS SageMaker.
- Built a framework for fitting custom scikit-learn models, including polynomial regression and gradient-boosted machines.
- Reviewed learning-to-rank methods and proposed applications to product recommendations.

Amazon

May 2020–Aug 2020

Software Development Engineer Intern

Seattle, WA

- Designed a recommender system for the Amazon Advertiser Partner Network (AAPN) and identified AWS tools for its implementation.
- Performed exploratory data analysis on the advertiser dataset.

Competitive Programming Achievements

- **ICPC World Finalist.** Placed 6th in the 2022 ICPC North American Championship and competed in the World Finals in Dhaka, Bangladesh.
- **Meta Hacker Cup, Top 200 (2022).** Placed in the top 200 in Round 3, out of more than 27,000 contestants worldwide.

Teaching Experience

Purdue University Jan 2024–Present
Computer Science Instructor

- Teach CS 311 (Competitive Programming), an intermediate-level competitive programming course.

X-Camp Academy Jan 2023–Present
USA Computing Olympiad Instructor

- Coach students at the USACO Silver and Gold levels.
- Serve as lead instructor for 40x/50x classes (USACO Gold Foundations).

Georgia Institute of Technology Jan 2021–Dec 2022
Teaching Assistant, Automata and Complexity (CS 4510)

- Served as a teaching assistant for Zvi Galil for four semesters.

Georgia Institute of Technology Aug 2019–Dec 2019
Teaching Assistant, Algorithms (CS 3510)

- Served as a teaching assistant for Eric Vigoda.

Leadership and Service

Georgia Tech Programming Team Jan 2022–Dec 2022
President

- Prepared content and delivered weekly lectures on dynamic programming, data structures, and combinatorial optimization.
- Organized team practices for ICPC.

Big O Theory Club Aug 2020–May 2021
President, Georgia Institute of Technology

- Delivered talks on theoretical computer science, including randomized algorithms and flows, and invited students to present their research.

Additional Service

- Reviewer: NeurIPS (2026); SODA (2027).
- Coach, Georgia ARML (American Regions Mathematics League) team, May–Jun 2022.
- Reviewer, Georgia Tech President's Undergraduate Research Awards, Fall 2022.

Technical Skills

Programming: Python, C++

Tools and platforms: PyTorch, scikit-learn, AWS SageMaker, AWS, Azure

Areas: Algorithm design, machine learning, optimization

Coursework and Certifications

Mathematics: Advanced Graph Theory, Graduate Probability I, Analysis I, Abstract Algebra I, Algebraic Topology I, High-Dimensional Statistics.

Computer science: Machine Learning Theory, Dynamic Algebraic Algorithms, High-Performance Computing, Deep Learning, Robot Intelligence: Planning, Applied Cryptography, Introduction to Database Systems, Natural Language Processing.

Certification: Coursera Deep Learning Specialization.